

SEVERO-KURILSK, Russian Federation

WMO: 322150

Lat: 50.6663N

Lon: 156.1189E

Elev: 23

StdP: 101.05

Time Zone: 11.00 (E11)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.6	-9.2	-16.6	0.9	-9.2	-14.8	1.0	-7.8	17.8	-2.9	15.9	-2.6	4.5	340	0.761

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	18.3	12.6	16.5	12.2	15.0	11.7	13.8	16.2	13.0	15.0	12.3	14.1	4.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	9.2	14.5	12.0	8.7	13.6	11.3	8.3	12.9	38.8	16.4	36.6	15.1	34.9	14.3	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.4	9.6	DB	-11.7	22.0	2.1	1.6	-13.2	23.2	-14.5	24.1	-15.6	25.0	-17.2	26.1	(4)
(5)			WB	-12.7	16.1	1.8	1.4	-14.1	17.0	-15.1	17.8	-16.2	18.6	-17.5	19.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-3.7	-4.8	-3.1	-0.3	3.0	7.1	10.3	11.8	10.6	6.1	1.2	-2.3
	DBStd	6.20	2.62	3.04	2.47	1.79	2.09	2.48	2.14	1.69	1.82	2.21	2.45	2.45
	HDD10.0	2668	425	413	405	308	218	93	22	3	13	120	266	381
	HDD18.3	5581	683	647	663	558	476	337	250	202	232	378	516	639
	CDD10.0	130	0	0	0	0	1	7	31	60	32	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.7	4.7	5.1	4.9	4.3	3.1	2.4	2.1	2.3	3.2	3.9	4.2	4.5
Precipitation	PrecAvg	1759	121	124	134	115	99	107	159	144	164	223	220	161
	PrecMax	2244	209	269	288	441	217	280	362	364	331	372	392	300
	PrecMin	1346	61	43	52	39	16	5	0	24	68	107	103	73
	PrecStd	215	38	60	54	75	48	73	109	91	61	71	75	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	2.1	3.0	8.1	13.9	20.1	20.9	20.1	18.4	12.7	8.2	4.6
		MCWB	0.8	0.4	1.1	4.4	7.3	11.5	12.7	13.7	12.4	9.0	6.6	3.2
	2%	DB	1.6	1.0	1.4	4.0	10.3	16.3	17.9	18.0	16.1	11.3	6.8	3.5
		MCWB	0.2	-0.2	-0.2	1.2	5.8	10.2	12.7	13.1	11.6	8.2	5.4	2.2
	5%	DB	0.7	0.3	0.7	2.6	7.7	13.6	15.5	16.3	14.9	10.4	5.9	2.4
		MCWB	-0.5	-0.9	-0.6	0.6	4.7	8.9	12.3	12.8	11.3	7.8	4.5	1.1
	10%	DB	0.1	-0.5	0.1	1.8	6.0	11.2	13.7	14.9	13.8	9.5	4.8	1.4
		MCWB	-1.1	-1.7	-1.0	0.2	3.7	7.9	11.5	12.2	10.7	7.1	3.2	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.0	1.2	4.7	8.9	12.5	15.1	15.5	13.9	10.1	7.1	3.8
		MCDB	2.1	1.5	2.0	7.9	11.9	17.9	18.4	18.4	15.7	11.3	7.8	4.3
	2%	WB	0.5	0.1	0.3	1.8	6.2	10.8	13.4	14.1	12.7	9.1	5.8	2.4
		MCDB	1.2	0.8	0.9	3.0	8.8	15.5	16.1	16.5	14.8	10.5	6.6	3.3
	5%	WB	-0.2	-0.7	-0.1	1.1	5.0	9.4	12.4	13.3	12.0	8.2	4.8	1.3
		MCDB	0.6	0.1	0.4	2.0	7.3	12.7	14.7	15.2	13.9	9.7	5.7	2.3
	10%	WB	-1.0	-1.6	-0.7	0.6	4.0	8.4	11.6	12.6	11.3	7.5	3.5	0.1
		MCDB	0.0	-0.7	0.0	1.5	5.6	10.7	13.6	14.3	13.0	9.1	4.6	1.2
Mean Daily Temperature Range	5% DB	MDBR	3.0	3.1	3.2	3.1	3.9	5.4	4.9	4.9	4.9	3.9	3.3	3.0
		MCDBR	3.2	3.4	3.5	4.6	7.7	10.8	9.2	8.2	7.4	5.2	4.7	3.6
		MCWBR	3.1	3.3	3.4	3.4	4.6	5.9	5.6	5.1	4.8	4.0	4.5	3.6
	5% WB	MCDBR	3.4	3.4	3.2	3.9	6.9	8.9	7.5	6.6	6.3	4.8	4.6	3.5
		MCWBR	3.6	3.6	3.3	3.2	4.5	5.5	5.1	4.6	4.5	4.2	4.8	3.8
Clear-Sky Solar Irradiance	taub	0.337	0.364	0.412	0.463	0.488	0.464	0.459	0.435	0.380	0.395	0.375	0.341	
	taud	2.114	2.067	1.972	1.896	1.934	2.099	2.177	2.251	2.376	2.150	2.078	2.076	
	Ebn at Noon	643	732	764	771	773	799	796	796	807	689	581	562	
	Edh at Noon	81	111	146	177	178	152	139	123	98	101	84	73	
All-Sky Solar Radiation	RadAvg	0.83	1.47	2.62	3.67	4.23	4.54	4.18	3.66	3.12	1.93	0.98	0.64	
	RadStd	0.06	0.15	0.22	0.38	0.47	0.53	0.45	0.35	0.23	0.14	0.08	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page